

Lecture 1

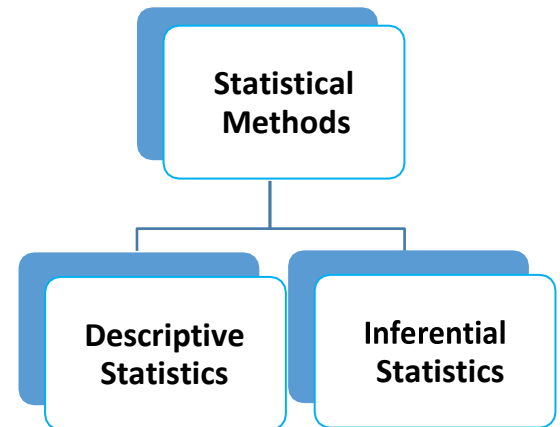
1- The Nature of Probability and Statistics

1-1 Descriptive and Inferential Statistics

Statistics is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.

Data can be used in different ways. **Statistics** is divided into two main areas, depending on how data are used:

- 1- **Descriptive statistics** consists of the collection, organization, presentation, and summarization of data.
- 2- **Inferential statistics** consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.



- A **population** consists of all subjects (human or otherwise) that are being studied.
- A **sample** is a group of subjects selected from a population.

Types of Descriptive Statistics:

1- *Measure of central tendency:*

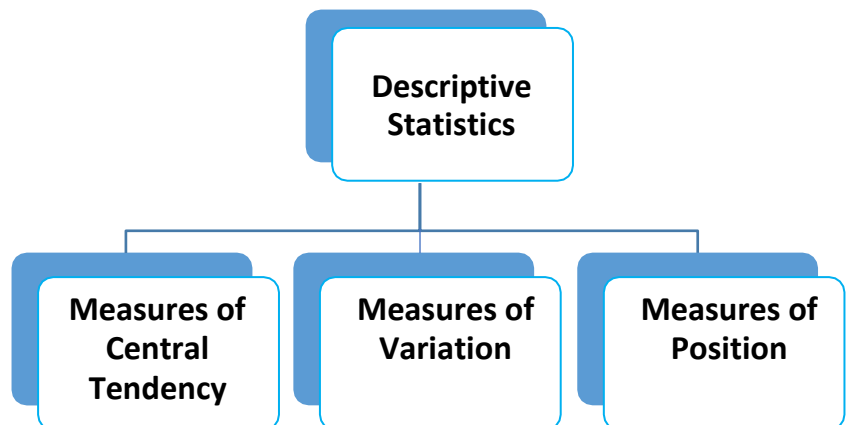
Mean, Median, Mode,
Midrange and Weighted Mean

2- *Measure of variation:*

Range, Variance, Standard
Deviation and Coefficient of Variation

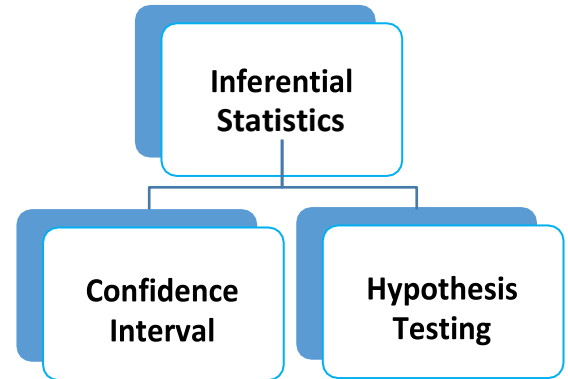
3- *Measure of position:*

Standard Score, Percentiles, Quartiles, Deciles and Outliers.



There are two major divisions of inferential statistics:

- 1- A *confidence interval* gives a range of values for an unknown parameter of the population by measuring a statistical sample.
- 2- Tests of significance or *hypothesis testing* where scientists make a claim about the population by analyzing a statistical sample.



The data is summarized by the researcher, in a useful way, with the help of numerical and graphical tools such as charts, tables, and graphs, to represent data in an accurate way. Moreover, the text is presented in support of the diagrams, to explain what they represent.

1-2 Variables and Types of Data

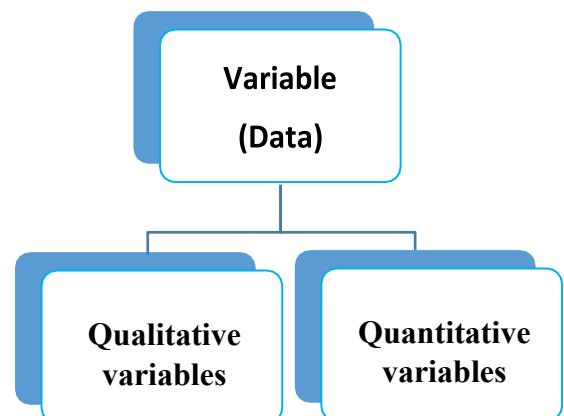
Data are the values (measurements or observations) that the variables can assume.

Different Types of Data:

A **variable** can be defined as a characteristic that can take different values (e.g. measures of temperature, height, weight, cost and gender) unlike **Constants** which are characteristics that have values that do not change (e.g. π (π) = 3.14159... and e = 2.71828....)

Variables can be classified as qualitative or quantitative:

- 1- **Qualitative variables** are variables that can be expressed not in terms of numbers, but rather by means of a natural language description, according to some characteristic or attribute.
For example: gender (male or female), beauty of individual, intelligence of people, ...



Qualitative variables can be further classified into two groups: **ordinal** and **nominal**:

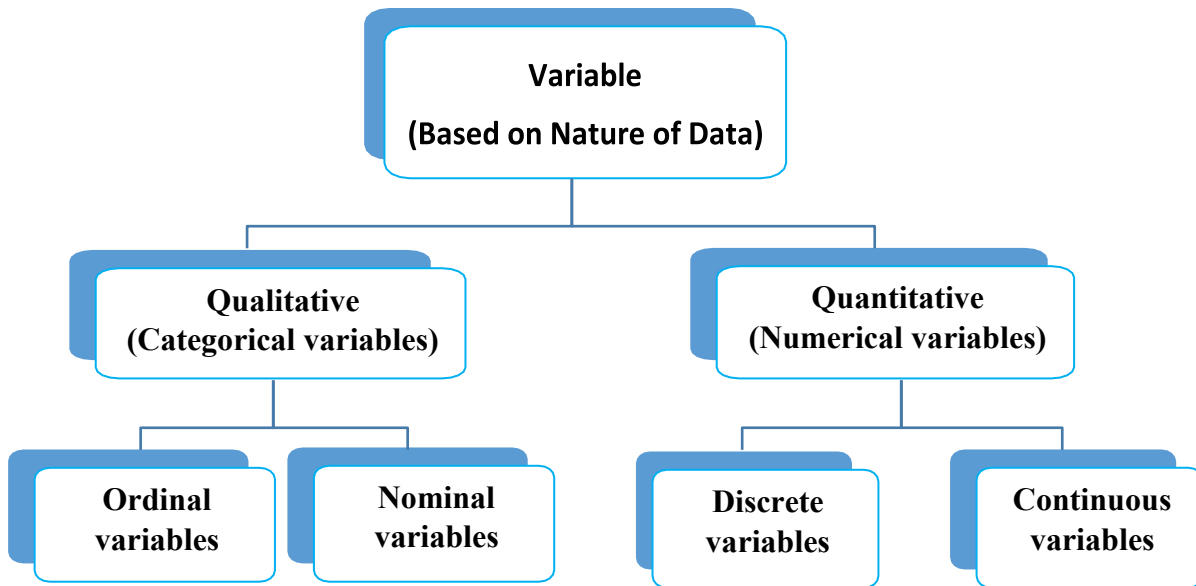
- i) A categorical variable whose categories can be meaningfully ordered is called **ordinal** variable. For example: size (small, medium, large, etc.), Attitudes (strongly disagree, disagree, neutral, agree, strongly agree), a student's grade in an exam (A, B, C or Fail) are ordinal variables.
- ii) When it does not matter which way the categories are ordered in tabular or graphical displays of the data (i.e. all orderings are equally meaningful), we call them **nominal** categories. For example: a student's religion (Atheist, Christian, Muslim, Hindu), gender, or sport.

2- Quantitative variables are numerical and can be ordered or ranked. For example, the variable *age* is numerical, and people can be ranked in order according to the value of their ages. Other examples of quantitative variables are heights, weights, and body temperatures.

Quantitative variables can be further classified into two groups: **discrete** and **continuous**:

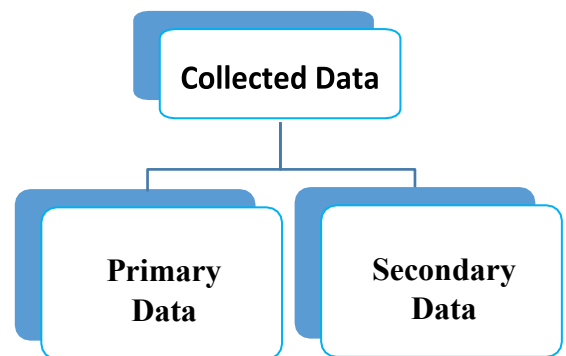
- i) A variable whose values are generally countable is called **discrete**. For example: the number of books, marks in examination.
- ii) A variable that may contain any value within some range is called **continuous**. It can take integral and fractional values both. For example: height, weight, length, time, etc.

So, in addition to being classified as qualitative or quantitative, variables can be classified by how they are categorized, counted, or measured as in the following chart:



1-3 Data Collection and Sampling Techniques

The method of collecting data is considered one of the most important phases that statistical research mainly depends on. Collecting data in such scientific methods will lead to get accurate outcomes when doing operations of statistical inference and proper decision making.



Resources of collecting data

1- Primary resources (Field resources):

These are the resources which we originally get data such as interviewing or questionnaires. This type is distinguished by accuracy. However, it needs time and efforts beside it is highly expensive to conduct such a type.

2- Secondary resources (*historical resources*):

We can get our data from authorities and agencies formally work such as central agency for public mobilization and statistics, internet and media. This type is a good type of resources such that it saves time and money.

Methods of collecting data

The method of collecting data is determined according to the aim and the size of the statistical society under study.

For example: The students of a faculty represent a statistical society whose value is the student.

1. *Method of mass population:*

It means to collect the data related to the phenomenon of the statistical society. It's used to include all the society such as the population. This type is including all the values. It's unbiased and the outcomes are so accurate.

The disadvantages of such a method are: it needs long time and great efforts. Furthermore, it costs much money.

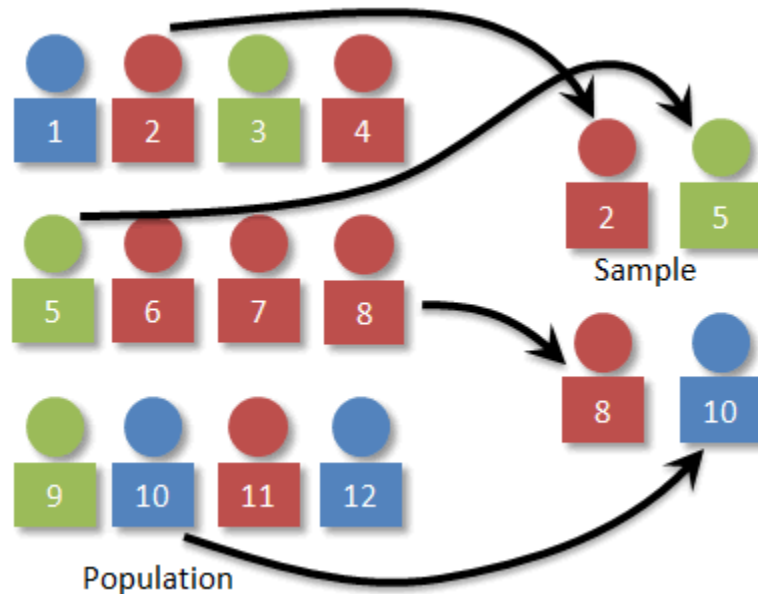
2. *Methods of samples:*

It mainly depends upon selecting a sample from the statistical society that it represents. We conduct researches on the sample. The outcomes we get are generalized on the whole society. This type saves time, efforts and money, ...

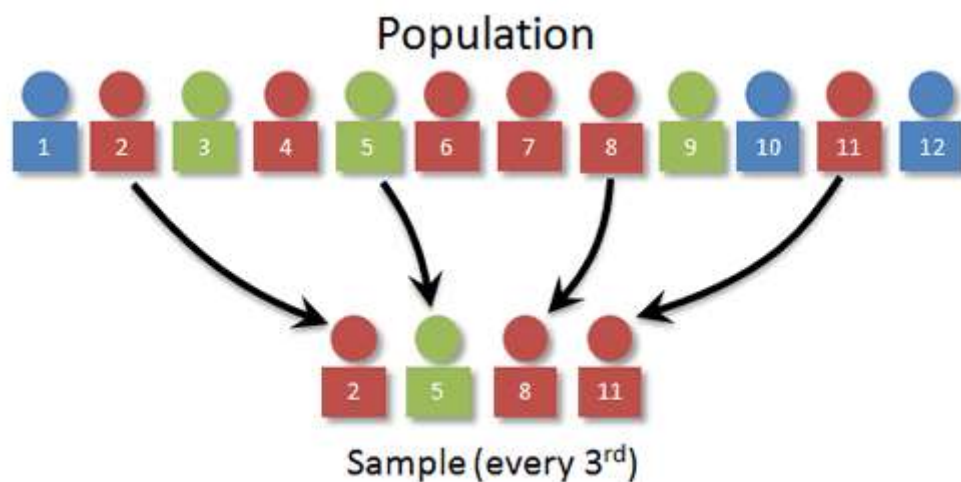
The disadvantage of the sample methods is: the outcomes of such type are not accurate if the selected sample doesn't represent all the society well. In such a case the sample is called biased.

To obtain samples that are unbiased—i.e., that give each subject in the population an equally likely chance of being selected—statisticians use four basic methods of sampling: **random**, **systematic**, **stratified**, and **cluster** sampling:

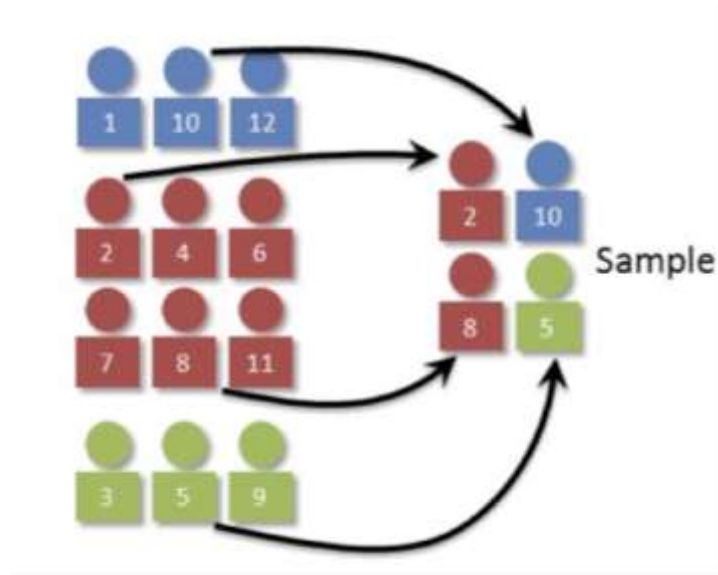
- **Random** Subjects are selected by random numbers.



- **Systematic** Subjects are selected by using every k^{th} number after the first subject is randomly selected from 1 through k .



- **Stratified** Subjects are selected by dividing up the population into groups (strata), and subjects are randomly selected within groups.



- **Cluster** Subjects are selected by using an intact group that is representative of the population.

